

Debosmita (Deb) Pathak

NSF Graduate Research Fellow

Dean's Distinguished University Fellow, Ohio State University

IPAC Visiting Graduate Student Research Fellow, IPAC/Caltech

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Main Interests: ✨ Stellar Feedback ✨ Polycyclic Aromatic Hydrocarbons & Dust ✨ Star Formation
🌀 Galaxy Evolution ☁ Interstellar Medium ✨ Multi-Wavelength Observations

Education

Ph.D., Astronomy , The Ohio State University, Columbus, OH	2022 – 2027 (Expected)
M.S., Astronomy , The Ohio State University, Columbus, OH	2022 – 2024
B.A. with Honors, Physics & Mathematics , Grinnell College, Grinnell, IA	2018 – 2022

Research Experience

◦ IPAC Visiting Graduate Student Research Fellow <i>Infrared Processing and Analysis Center, California Institute of Technology</i> Advisors: Dr. Lee Armus, Dr. Laura Lenkic, Dr. Thomas Lai, Dr. Vivian U	Pasadena, CA Feb 2026 – Aug 2026
◦ UCSD Visiting Scholar <i>University of California, San Diego</i> Advisor: Dr. Karin M. Sandstrom	San Diego, CA Sept 2024 – Dec 2024
◦ NSF Graduate Research Fellow & Dean's Distinguished University Fellow <i>The Ohio State University</i> Advisors: Dr. Adam K. Leroy, Dr. Todd A. Thompson, Dr. Laura A. Lopez	Columbus, OH Aug 2022 – Present
◦ Undergraduate Researcher <i>Grinnell College</i> Advisor: Dr. Charlotte R. Christensen	Grinnell, IA 2021 – 2022
◦ Undergraduate Researcher <i>Harvard-Smithsonian Center for Astrophysics</i> Advisors: Dr. Sirio Belli, Dr. Rainer Weinberger	(virtual) Boston, MA 2020 – 2021

Publications

6 as first author, 18 as contributing author, 1 in prep. [🔗](#)

1. **Pathak, D.** et al. (in prep.). Dusty and Over-Pressured: Radiation Pressure Drives the Evolution of Young, Massive Star Clusters in NGC 3256 (to be submitted to *ApJL*)
2. **Pathak, D.** et al. (2026). Mid-Infrared Colors Vary with Galactic Environment: Contrasting Star-Forming Disks, Young Centers, and Quiescent Star-Formation Deserts (*ApJ*)
3. **Pathak, D.** et al. (2025). Masses, Star-Formation Efficiencies, and Dynamical Evolution of 18,000 HII Regions (*ApJL*)
4. **Pathak, D.** et al. (2025). Survivors and Zombies: The Quenching and Disruption of Satellites Around Milky Way Analogs (*ApJ*)
5. **Pathak, D.** et al. (2025). Linking Stellar Populations to HII Regions across Nearby Galaxies. II. Infrared Reprocessed and UV Direct Radiation Pressure in HII Regions (*ApJ*)
6. **Pathak, D.** et al. (2024). A Two-Component Probability Distribution Function Describes the Mid-IR Emission from the Disks of Star-forming Galaxies (*AJ*)
7. **Pathak, D.** et al. (2021). Quenching, Mergers, and Age Profiles for $z = 2$ Galaxies in IllustrisTNG (*ApJL*)

Observing Proposals

PI: Pathak, JWST GO 10350; Dust in the Deserts: Understanding PAH Emission in Quiescent Bulges and Star-formation Deserts (60.6 hours, \$383,652.00)

PI: Kim (incl. **co-I Pathak**), **JWST GO 11148; Where the PAH-CO Relation Breaks: JWST Spectroscopy of High CO/PAH Ridges in NGC 1566**

PI: Leroy (incl. **co-I Pathak**), **JWST GO 10046; Resolving Stellar Feedback in Action via NIRCам and MIRI Imaging of Very Nearby Galaxies**

PI: Linden (incl. **co-I Pathak**), **HST GO 18400; Under Pressure: Star Cluster Formation and Evolution in Merger-Induced Starburst Galaxies**

PI: Leroy (incl. **co-I Pathak**), **JWST GO 3707; A JWST Census of the Local Galaxy Population: Anchoring the Physics of the Matter Cycle**

PI: Leroy (incl. **co-I Pathak**), **JWST GO 2987; Resolving HII Regions and ISM Structure Across the Milky Way Analog NGC 253**

PI: Wang (incl. **co-I Pathak**), **ALMA 2026.1.00428.S; First spatially resolved CO-PAH study in z=1 main sequence galaxies: ALMA CO follow-up on JWST IFU detected PAHs**

PI: Leroy (incl. **co-I Pathak**), **ALMA 2025.1.00576.L; The 10 pc Survey of Molecular Clouds and Stellar Feedback**

Honors & Awards

IPAC Visiting Graduate Student Research Fellowship, IPAC	2026
Allan H. Markowitz Award in Observational Astronomy, OSU	2025
AAS International Travel Grant, IAU General Assembly	2024
AAS Chambliss Honorable Mention	2024
NSF Graduate Research Fellowship, OSU	2022–Present
Susan Huntington Dean’s Distinguished University Fellowship, OSU	2022–Present
H. George Apostle Prize in Physics, Grinnell College	2022
Phi Beta Kappa Honors, Grinnell College	2022
Leopold Schepp Foundation Scholarship, Grinnell College (\$4,000)	2021–2022
AAS Chambliss Honorable Mention	2021

Leadership & Service

Member; PHANGS Colloquium Organizing Committee	2023–Present
Judge; OSU Graduate Ray Travel Award (3 application cycles/year)	2023–Present
Volunteer; IPAC Events at AAS246 & Pasadena Day of Astronomy, Pasadena, CA	Summer 2026
Chair; Health, Wellness & Safety Committee, OSU Council of Graduate Students	2024–2026
Volunteer; Boston Pride Parade For The People, Boston, MA	Summer 2026
Judge; OSU Council of Graduate Students Distinguished Service Award	2026
Organizer; Klondike Kolloquium (OSU Graduate Student Summer colloquium series)	2023–Present
Co-Chair & Organizer; Annual Buckeye Accessibility Stakeholders’ Symposium, OSU	2024–2025
Volunteer; Graduate Student Appreciation Week	2024–2025
Senator; Ohio State University Senate	2023–2025
Elected Member; Graduate Caucus Committee, OSU Senate	2023–2025
Elected Member; OSU Council of Graduate Students Executive Committee	2023–2026
Elected Delegate for Department of Astronomy, OSU Council of Graduate Students	2022–2026
Appointed Member; Graduate Compensation & Benefits Committee, OSU	2022–2025
Mentor; Applying for Graduate Schools, Career, Life & Services, Grinnell College	2024
Appointed Member; Council of Academic Affairs, OSU	2022–2024
Appointed Member; Senate Diversity Committee, OSU	2023–2025
Instructor; Undergraduate Summer Python Bootcamp, OSU	2023–2025
Mentor; Undergraduate Group Project (exploring PHANGS data; 15 mentees), OSU	2023
Mentor; URSA Summer Program (3 mentees), OSU	Summer 2023
Instructor/Academic Facilitator; URSA Summer Program (14 students), OSU	Summer 2023
Graduate Member; Astronomy Department Diversity Committee, OSU	2022–2025
Mentor; Polaris Undergraduate Mentorship Program, OSU	2022

Presentations

14 *Invited Talks*, 1 *Press Conference*, 11 *Contributed Talks*, 8 *Poster Presentations*

Invited Talk , Galaxies Seminar, University of Michigan, Ann Arbor, MI	Oct 2026
Invited Talk , Astronomy Seminar, University of Kansas, Lawrence, KS	Oct 2026
Invited Talk , AstroCoffee, Johns Hopkins University, Baltimore, MD	Oct 2026
Invited Talk , Center for Theory and Computation Seminar, University of Maryland, College Park, MD	Sept 2026
Invited Talk , SF/ISM Seminar, Princeton University, Princeton, NJ	Sept 2026
Invited Talk , Galaxy Seminar, Center for Computational Astrophysics, NYC, NY	Sept 2026
Invited Talk , Lunch Seminar, Columbia University, NYC, NY	Sept 2026
Invited Talk , IPAC Lunch Seminar, IPAC/Caltech, Pasadena, CA	Jul 2026
Press Conference , AAS 249, Pasadena, CA	Jun 2026
Talk , AAS 249, Pasadena, CA	Jun 2026
Talk , GOALS Team Meeting, Pasadena, CA	Jun 2026
Invited Talk , PHANGS & Friends Meeting, McMaster University, Hamilton, Canada	May 2026
Invited Talk , Astronomy Seminar, University of Kentucky, Lexington, KY	Apr 2026
Invited Talk , SDSS Science Friday, virtual	Nov 2025
Poster , Pan Dust Conference, University of Arizona, Tucson, AZ	Nov 2025
Talk , Stellar Feedback Conference, Ascona, Switzerland	Aug 2025
Poster , 18th Potsdam Thinkshop on Feedback, Potsdam, Germany	Jul 2025
Talk , Galactic Systems Under the Microscope, ESO, Garching, Munich, Germany	Jul 2025
Talk , 2025 STScI Spring Symposium, STScI, Baltimore, MD	May 2025
Invited Talk , Galaxy Journal Club, STScI, Baltimore, MD	May 2025
Talk , Interstellar Medium* Group (ISM*@ST), STScI, Baltimore, MD	May 2025
Poster , Edward F. Hayes Advanced Research Forum, OSU, Columbus, OH	Mar 2025
Invited Talk , PHANGS Colloquium, virtual	Jan 2025
Talk , PAH Conference, University of Virginia, Charlottesville, VA	Nov 2024
Talk , NSBP-NSHP Joint Meeting, Houston, TX	Nov 2024
Invited Talk , Journal Club, University of California, San Diego, San Diego, CA	Oct 2024
Talk , Ensanada-San Diego Astrophysical Society Meeting, Ensanada, Mexico	Sept 2024
Talk , ISM Focus Meeting, IAU General Assembly, Cape Town, South Africa	Aug 2024
Poster , HI Symposium, IAU General Assembly, Cape Town, South Africa	Aug 2024
Poster , AAS 243, New Orleans, LA	Jan 2024
Poster , The First Year of JWST Science Conference, STScI, Baltimore, MD	Sept 2023
Talk , Olympian Symposium, Paralia Katerini, Thessaloniki, Greece	May 2023
Poster , AAS 241, Seattle, WA	Jan 2023
Poster , AAS 237, virtual	Jan 2021
(<i>Misc.</i>) Summer School , International Summer School on the ISM of Nearby Galaxies (GISM2), Banyuls-sur-Mer, France	Jul 2023

Software

Languages: Python, R, SQL, C++, C, JavaScript

Astronomy, Image Processing, and Machine Learning: astropy, scikit-learn, SLUG, pyneb, OpenCV, pandas, SciPy, DS9, CARTA, scikit-learn, UMAP, DOLPHOT, JWST Science Calibration pipeline (jwst), synphot, AstroDrizzle, CIGALE, Starburst99, CAFE, CRETA